Q. What are the various problem, which you have observed in the society & community. Pick any 5 problems. Find out if they can be solved with data structures and algorithms.

Q1.

i) Bus Seat Reservation: (Array)

It uses Array because no. of seats are fixed. Each seat is assigned seat number & stored at a specific place in an array. when we book or cancels a seat, the system directly access the location. Array help in displaying available and booked seats.

ii) Classroom Seating Arrangement (2D Array):

It is 2D array because seats are in rows and columns. Each element in the array stores information about student occupying that seat. It provides quick access to any seat based on its row and column.

iii) College Time Table (2D Array):

It is 2D array, where rows are days of week and columns are class periods. Arrays make it easy to display, update, and retrieve timetable information.

iv) ~~Student Roll no. List~~ Lift Management (Queue):

It is queue because in this queue data structure, passengers are served in the order they arrive. This follows First In, First Out. The first person waiting for the lift enters first, ensuring fairness and efficient movement of passengers.

v) ATM waiting line (Queue):

It is queue data structure. This follows FIFO. customers who arrive first are served first, while new ~~cus~~ customers join at the end of the line.